

Tutorial-3

Bhanu Prasad cs24b059

Kernel	Version	Cycles	Instructions	IPC	CPI
Matmul	Original	3,93,76,749	9,79,21,228	2.49	0.401
Matmul	Interchanged	3,40,46,375	10,75,26,671	3.16	0.316
Transpose	Original	1,08,76,808	1,15,66,030	1.06	0.943
Transpose	Interchanged	1,00,67,859	1,16,80,731	1.16	0.862

Table 1: Comparison between original and interchanged versions

Analysis Questions & Answers

1. How does loop interchange affect locality?

- **Matmul:** In the original (i-j-k), accessing $B[k * N + j]$ causes stride-N memory access, leading to poor spatial locality. In the interchanged (i-k-j), both $B[k * N + j]$ and $C[i * N + j]$ are accessed sequentially (stride-1), improving cache hits.
- **Transpose:** Interchanging loops only swaps which matrix has good locality. One matrix always suffers from stride-N access, so overall locality does not improve.

2. Which ordering improves cache behavior?

- **Matmul:** The (i-k-j) ordering improves cache utilization by accessing contiguous memory and reusing loaded cache lines effectively.
- **Transpose:** Neither ordering significantly improves cache performance. Techniques like blocking/tiling are required for better cache usage.

3. How does IPC change after transformation?

- **Matmul:** IPC increases because fewer cache misses reduce CPU stalls, allowing more instructions to execute per cycle.
- **Transpose:** IPC remains low since execution is dominated by memory access latency.

4. Does instruction count change significantly? Why?

- At -00: Instruction count is nearly the same since the total number of operations does not change.
- At -03: Interchanged matmul may enable SIMD vectorization, reducing instruction count by processing multiple elements per instruction.

5. Which kernel is more sensitive to loop ordering?

- **Matmul:** Highly sensitive; loop interchange greatly reduces cache misses and improves performance.
- **Transpose:** Less sensitive, as loop interchange cannot eliminate poor memory access patterns.

6. Does loop interchange change compute/memory behavior?

- **Matmul:** Moves toward compute-bound behavior as memory bottlenecks are reduced.
- **Transpose:** Remains memory-bound since it mainly involves data movement with little computation.